Imperial College London

Department of Electronic and Electrical Engineering

Examination 2026

Computational Sensing and Imaging (ELEC70085)

MSc and EEE/EIE PART IV

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All Questions are Compulsory

Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

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Questions [100 Marks]

Q1 **Forward Models for Coded Illumination, Coded Aperture, and Coded Sensor.** **[25 Marks]**

In computational imaging, one can encode information at 3 different stages of the imaging pipeline:

i) *coded illumination* (before light interacts with the scene),
ii) *coded aperture* (inside the optical system),
iii) *coded sensor* (at detection, e.g., per-pixel weighting).

Consider a 1D discrete scene $\mathbf{x} \in \mathbb{R}^N$, an optical blur modeled by convolution with a kernel $\mathbf{h}$ (i.e., $\mathbf{Hx} = \mathbf{h} * \mathbf{x}$), and sensor noise $\boldsymbol{\eta}$.

(a) **Coded illumination model.** Suppose we illuminate the scene with a known illumination pattern $\mathbf{m} \in \mathbb{R}^N$ (e.g., from a projector), so that the effective scene becomes $\mathbf{m} \odot \mathbf{x}$ (pointwise multiplication). Write the forward model for the measurements $\mathbf{y}_{\text{illum}}$ in terms of $\mathbf{m}$, $\mathbf{H}$, and $\mathbf{x}$. Briefly explain (in 1-2 sentences) how coding *before the scene* affects the information captured. **[4 Marks]**

Under coded illumination, the scene becomes $\mathbf{m} \odot \mathbf{x}$, so

$$\mathbf{y}_{\text{illum}} = \mathbf{H}(\mathbf{m} \odot \mathbf{x}) + \boldsymbol{\eta}.$$

Since coding happens *before* the scene, the measurement encodes portions of $\mathbf{x}$ according to where light is injected; this can improve conditioning or enable multiplexing even when the optics and sensor are simple.

(b) **Coded aperture model.** Assume uniform illumination but insert a known mask $\mathbf{a}$ in the optical path, modifying the effective point spread function (PSF) to $\mathbf{h}_{\text{eff}} = \mathbf{a} * \mathbf{h}$. Write the forward model $\mathbf{y}_{\text{ap}}$ and explain briefly how aperture coding differs conceptually from illumination coding. **[4 Marks]**

With a mask $\mathbf{a}$ modifying the optical transfer, the forward model is

$$\mathbf{y}_{\text{ap}} = (\mathbf{a} * \mathbf{h}) * \mathbf{x} + \boldsymbol{\eta} = \mathbf{H}_{\text{eff}}\mathbf{x} + \boldsymbol{\eta}.$$

Here the coding acts *in the optics*, shaping the blur/PSF seen by every point of the scene. Unlike illumination coding, it does not inject new light but reshapes how the scene is mixed during imaging.

(c) **Coded sensor model.** Finally, assume illumination and optics are standard, but the sensor applies per-pixel gains collected into a diagonal matrix $\mathbf{W} = \text{diag}(w_1, \ldots, w_N)$. Write the forward model $\mathbf{y}_{\text{sens}}$, and explain briefly what is fundamentally different about coding at the sensor compared to coding at illumination and aperture. **[4 Marks]**

The coded sensor model is

$$\mathbf{y}_{\text{sens}} = \mathbf{W}(\mathbf{Hx}) + \boldsymbol{\eta}.$$

Here coding happens *after* optics, at the moment of detection. The sensor weights or mixes what the optics deliver, without altering how the scene is illuminated or how the optics blur it.

(d) **Comparison.** Summarize in a simple table or 3–4 sentences the key differences among coded illumination, coded aperture, and coded sensor in terms of:

- where coding occurs,
- what part of the physics it modifies,
- and what flexibility or limitations it introduces. **[3 Marks]**

A concise comparison:

- *Coded illumination:* coding occurs *before* the scene; it modulates the photon field incident on $\mathbf{x}$, enabling active probing but requiring a programmable light source.
- *Coded aperture:* coding occurs *in the optics*; it reshapes the PSF and encodes spatial/angular/spectral structure but reduces light throughput.
- *Coded sensor:* coding occurs *at detection*; the optics remain unchanged while per-pixel weights or filters encode the signal, offering compact hardware but limited flexibility once fabricated.

(e) **(Mix-and-match) Classifying real-world coded imaging systems (Illumination / Aperture / Sensor).** **[10 Marks]**

Below is a list of real-world imaging hardware systems discussed in the class. Each one uses coding at a specific stage of the imaging pipeline: illumination, optics, or sensor.

Match each system (1)-(10) below to exactly one of the following three categories:

I] coded illumination
A] coded aperture (coding in the optics)
S] coded sensor

1) Per-pixel coded exposure (flutter shutters on a CMOS sensor)
2) DMD-based structured light projection (e.g., binary/Gray/Hadamard patterns)
3) MURA / URA mask for high-energy (X-ray/gamma) imaging
4) Time-of-flight (ToF) sensors with per-pixel electronic modulation
5) Phase-mask wavefront coding (e.g., cubic-phase mask for extended depth-of-field)
6) Color filter arrays (e.g., Bayer, Quad-Bayer, multispectral mosaics)
7) Light-field / plenoptic camera with microlens array at an aperture conjugate plane
8) Compressive cameras with in-pixel mixing (e.g., ASPs, compressive CMOS arrays)
9) Rolling-shutter spatiotemporal coding (row-wise phase shifts)
10) Coded flash / temporally modulated illumination for motion deblurring)

- (1) Per-pixel coded exposure / flutter shutters $\rightarrow$ (S)
- (2) DMD structured light projection $\rightarrow$ (I)
- (3) MURA / URA mask (high-energy imaging) $\rightarrow$ (A)
- (4) Time-of-flight per-pixel modulation $\rightarrow$ (S)
- (5) Phase-mask wavefront coding (cubic phase) $\rightarrow$ (A)
- (6) Color filter arrays $\rightarrow$ (S)
- (7) Plenoptic microlens array $\rightarrow$ (A)
- (8) In-pixel mixing / compressive CMOS $\rightarrow$ (S)
- (9) Rolling-shutter spatiotemporal coding $\rightarrow$ (S)
- (10) Coded flash / temporal illumination modulation $\rightarrow$ (I)

Rationale. Items (1), (4), (6), (8), (9) encode information at the sensor via per-pixel filtering, modulation, timing, or in-pixel mixing. Items (2) and (10) actively encode the light projected onto the scene (illumination coding). Items (3), (5), and (7) place a mask or engineered optic into the optical path, modifying the PSF before detection (aperture/optics coding).

Q2 **Constrained Least Squares (CLS) for a toy imaging system.** **[20 Marks]**

Consider a simplified (toy) 2-pixel imaging system with an unknown scene vector $\mathbf{x} \in \mathbb{R}^2$ (two reflectivities). The optics/sensor pipeline is modeled as a 2×2 linear operator $\mathbf{H}$ (blur/crosstalk) followed by a per-pixel gain matrix $\mathbf{W}$ (pixel responsivity). The measurement model is

$$\mathbf{y} = \mathbf{WHx} + \boldsymbol{\eta},$$

where $\boldsymbol{\eta}$ is measurement noise.

We pose the following constrained least-squares (CLS) problem:

$$\min_{\mathbf{x}\in\mathbb{R}^2} \left\|\mathbf{WHx} - \mathbf{y}\right\|_2^2, \qquad \text{subject to} \quad \mathbf{Cx} = \mathbf{b},$$

with

$$\mathbf{W} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Interpretation: $\mathbf{Cx} = \mathbf{b}$ encodes simple scene priors/constraints (e.g., known total brightness and left-right symmetry).

(a) Write down the Lagrangian $\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda})$. **[4 Marks]**

Imaging setup. Define the forward operator

$$\mathbf{A} \coloneqq \mathbf{WH} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix},$$

so that the predicted measurements are $\widehat{\mathbf{y}} = \mathbf{Ax}$ and the data-fit term is $\|\mathbf{Ax} - \mathbf{y}\|_2^2$.

Lagrangian construction. To incorporate the equality constraints $\mathbf{Cx} = \mathbf{b}$, introduce Lagrange multipliers $\boldsymbol{\lambda} \in \mathbb{R}^2$ and form

$$\boxed{\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}) = \|\mathbf{Ax} - \mathbf{y}\|_2^2 + \boldsymbol{\lambda}^\top(\mathbf{Cx} - \mathbf{b})}.$$

This augments the least-squares objective with linear terms that enforce feasibility at optimality.

(b) Derive the optimality conditions in matrix form (a linear system in $(\mathbf{x}, \boldsymbol{\lambda})$). **[6 Marks]**

Stationarity with respect to $\mathbf{x}$.

$$\nabla_{\mathbf{x}}\mathcal{L} = 2\mathbf{A}^\top(\mathbf{Ax} - \mathbf{y}) + \mathbf{C}^\top\boldsymbol{\lambda} = \mathbf{0}.$$

Primal feasibility.

$$\mathbf{Cx} = \mathbf{b}.$$

KKT linear system. Stacking these gives the saddle-point system in $(\mathbf{x}, \boldsymbol{\lambda})$:

$$\boxed{\begin{bmatrix} 2\mathbf{A}^\top\mathbf{A} & \mathbf{C}^\top \\ \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix} = \begin{bmatrix} 2\mathbf{A}^\top\mathbf{y} \\ \mathbf{b} \end{bmatrix}}.$$

Explicit matrices for this imaging problem. With $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ we have

$$\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix}, \qquad \mathbf{A}^\top\mathbf{y} = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}.$$

Therefore,

$$2\mathbf{A}^\top\mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 10 \end{bmatrix}, \qquad 2\mathbf{A}^\top\mathbf{y} = \begin{bmatrix} 6 \\ 10 \end{bmatrix}.$$

(These forms are useful for explicitly solving the 4×4 KKT system if desired.)

(c) Solve the optimal system in closed form to obtain the optimal $\mathbf{x}^\star$.

[6 Marks]

Key observation (imaging prior dominates). Here $\mathbf{C} \in \mathbb{R}^{2\times 2}$ is invertible. Hence the constraints (which encode the scene priors: total brightness and symmetry) uniquely determine $\mathbf{x}$; there are no free degrees of freedom left for the LS term to further optimize.

Solve the constraints.

$$\mathbf{C}\mathbf{x} = \mathbf{b} \iff \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \iff \begin{cases} x_1 + x_2 = 1, \\ x_1 - x_2 = 0. \end{cases}$$

Adding the equations gives $2x_1 = 1 \Rightarrow x_1 = \frac{1}{2}$, and then $x_2 = x_1 = \frac{1}{2}$.

Optimal scene (reflectivities).

$$\boxed{\mathbf{x}^\star = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}.}$$

Interpretation in the imaging context. The prior "total brightness $=$ 1" and "left-right symmetry" together imply an equal split of brightness across the two pixels.

(d) Verify that $\mathbf{C}\mathbf{x}^\star = \mathbf{b}$, and report the achieved residual $\|\mathbf{W}\mathbf{H}\mathbf{x}^\star - \mathbf{y}\|_2^2$. **[4 Marks]**

Constraint check (feasibility).

$$\mathbf{C}\mathbf{x}^\star = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{b}.$$

Forward model and residual in the sensor domain.

$$\mathbf{A}\mathbf{x}^\star = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{r}^\star := \mathbf{A}\mathbf{x}^\star - \mathbf{y} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Therefore,

$$\boxed{\|\mathbf{W}\mathbf{H}\mathbf{x}^\star - \mathbf{y}\|_2^2 = \|\mathbf{r}^\star\|_2^2 = 1.}$$

Imaging takeaway. Because the priors (constraints) fully fix the scene, the data-fit cannot be further improved; the estimator must accept the best feasible scene consistent with the priors, which in this case yields residual energy 1 at the sensor.

Q3 **One-Bit Time-Resolved Imaging.** **[15 Marks]**

Context. Time-resolved imaging systems (e.g., ToF cameras) attempt to measure the time-of-flight of a light pulse reflected from a scene. If a pulse $p(t)$ is emitted and the reflection is $r(t) = \Gamma p(t - t_0)$, then estimating the unknown delay t_0 reveals depth. However, sampling $r(t)$ at high temporal resolution (nanoseconds) across megapixel sensors lead to enormous data rates. To reduce data size, we explore combining *one-bit sampling* with time-resolved imaging.

The standard one-bit (first-order sigma–delta) scheme for a bandlimited signal $f(t)$ is:

$$\begin{cases} q[n] = \mathrm{sgn}(\, u[n-1] + f[n]\,), \\ u[n] = u[n-1] + f[n] - q[n], \end{cases}$$

where $q[n] \in \{-1, +1\}$ is the quantized output, $u[n]$ is the internal accumulator, and sgn is the sign or signum function.

(a) Explain why megapixel time-resolved imaging is difficult? **[3 Marks]**

A time-resolved sensor records not a single intensity but an entire *temporal waveform* at each pixel at nanosecond resolution. If the sensor has 10^6 pixels and each pixel records hundreds or thousands of time samples, the data volume grows as (number of pixels) $\times$ (temporal samples), leading to extremely large data rates and memory needs. Thus achieving megapixel spatial resolution together with fine temporal sampling is prohibitively expensive in bandwidth and storage.

(b) Let $p[n] = p(nT)$ be samples of a bandlimited pulse and let the reflected signal be

$$r[n] = \Gamma\, p(nT - t_0).$$

Using the one-bit model above, *express* $q[n]$ *using* $r[n]$ *and* $u[n]$ *alone*. **[3 Marks]**

Since $f[n] = r[n]$, the one-bit update becomes

$$q[n] = \mathrm{sgn}\big(u[n-1] + r[n]\big) = \mathrm{sgn}\big(u[n-1] + \Gamma p(nT - t_0)\big).$$

This is the compact expression relating the one-bit output to the reflected pulse.

(c) Using $q[n]$ from above and the linearity of the Discrete-Time Fourier Transform (DTFT), show that

$$\widehat{q}(\omega) \approx \Gamma e^{-j\omega t_0}\, \widehat{p}(\omega) - \widehat{u}(\omega)\, \widehat{v}(\omega),$$

for an appropriate filter $v[n]$. **[3 Marks]**

Hint: Use $u[n] - u[n-1] = r[n] - q[n]$*, i.e., a discrete derivative.*

From the update rule,

$$u[n] - u[n-1] = r[n] - q[n].$$

This is a discrete derivative:

$$(u * v)[n] = r[n] - q[n], \qquad v[n] = \delta[n] - \delta[n-1].$$

Taking the DTFT:

$$\widehat{u}(\omega)\widehat{v}(\omega) = \widehat{r}(\omega) - \widehat{q}(\omega).$$

Since $r[n] = \Gamma p(nT - t_0)$, its DTFT is $\widehat{r}(\omega) = \Gamma e^{-j\omega t_0}\widehat{p}(\omega)$, yielding

$$\widehat{q}(\omega) = \Gamma e^{-j\omega t_0}\widehat{p}(\omega) - \widehat{u}(\omega)\widehat{v}(\omega).$$

(d) Our goal is to estimate t_0 by matched filtering. For the original (unquantized) system,

$$t_0^\star = \arg\max_t |(r * \overline{p})(t)|, \qquad \overline{p}(t) = p(-t).$$

Show that the *same strategy* can be applied to $q[n]$ instead of $r[n]$ to estimate t_0.

[6 Marks]

Hints:

- Convolution with $\overline{p}$ acts as a *low-pass filter*.
- From part (c), $q = r - (u * v)$ where v is a discrete derivative.
- A derivative v corresponds to a high-pass filter in frequency.

We have

$$q[n] = r[n] - (u * v)[n].$$

The DTFT of $v[n] = \delta[n] - \delta[n-1]$ is $\widehat{v}(\omega) = 1 - e^{-j\omega}$, a high-pass function. Matched filtering with $\overline{p}$ corresponds to multiplication by $\widehat{\overline{p}}(\omega) = \overline{\widehat{p}(\omega)}$, which is low-pass because p is bandlimited.

Thus,

$$(q * \overline{p})(t) = (r * \overline{p})(t) - ((u * v) * \overline{p})(t).$$

But $(u * v)$ is high-pass, and convolving with the low-pass $\overline{p}$ strongly suppresses it:

$$((u * v) * \overline{p})(t) \approx 0.$$

Therefore,

$$(q * \overline{p})(t) \approx (r * \overline{p})(t),$$

which means t_0 can be estimated from q using the same matched-filter approach:

$$t_0^\star = \arg\max_t |(q * \overline{p})(t)|.$$

Q4 **From a Single Cosine to AR Modeling for Spectral Estimation.** **[20 Marks]**

Consider a real, noiseless discrete-time cosine

$$\boxed{x[n] = c_0 \cos(\Omega_0 n + \phi_0), \qquad n = 0, \ldots, N-1,}$$

with unknown amplitude $c_0 > 0$, frequency $\Omega_0 \in (0, \pi)$, and phase $\phi_0 \in (-\pi, \pi]$.

(a) Show that $x[n]$ satisfies a second-order autoregressive (AR(2)) recursion of the form

$$x[n] + a_1 x[n-1] + a_2 x[n-2] = 0, \qquad n = 2, \ldots, N-1,$$

and express a_1, a_2 in terms of Ω_0. **[4 Marks]**

Complex-exponential view. Write the cosine as a sum of two complex exponentials:

$$x[n] \;=\; \frac{c_0}{2} e^{j\phi_0} e^{j\Omega_0 n} \;+\; \frac{c_0}{2} e^{-j\phi_0} e^{-j\Omega_0 n} \;=\; d\, r^n + \overline{d}\, \overline{r}^{\,n},$$

where $r = e^{j\Omega_0}$ and $d = (c_0/2)e^{j\phi_0}$. Any sequence of the form $\sum_k d_k r_k^n$ with two distinct r_k satisfies a homogeneous linear recurrence whose characteristic polynomial has roots r and $\overline{r}$.

AR(2) recursion. Let the monic polynomial be $A(z) = 1 + a_1 z^{-1} + a_2 z^{-2}$ with roots at $z = r$ and $z = \overline{r}$. Then

$$A(z) \;=\; (1 - r z^{-1})(1 - \overline{r}\, z^{-1}) \;=\; 1 - (r + \overline{r}) z^{-1} + |r|^2 z^{-2}.$$

Since $|r| = 1$ and $r + \overline{r} = 2\cos\Omega_0$, we obtain

$$a_1 \;=\; -\,2\cos\Omega_0, \qquad a_2 \;=\; 1.$$

Thus $x[n]$ obeys $x[n] - 2\cos\Omega_0\, x[n-1] + x[n-2] = 0$, which is the stated AR(2) with $a_1 = -2\cos\Omega_0,\ a_2 = 1$.

(b) Using the recursion:

$$x[n] + a_1 x[n-1] + a_2 x[n-2] = 0, \qquad n = 2, \ldots, N-1,$$

form a linear system to estimate the AR parameters (a_1, a_2) from $x[n]$, and give the minimum value of N required in the noiseless case. **[6 Marks]**

Linear prediction form. For each $n = 2, \ldots, N-1$,

$$x[n] + a_1 x[n-1] + a_2 x[n-2] \;=\; 0 \;\Longleftrightarrow\; a_1 x[n-1] + a_2 x[n-2] \;=\; -x[n].$$

Stack the $N-2$ equations to obtain

$$\underbrace{\begin{bmatrix} x[1] & x[0] \\ x[2] & x[1] \\ \vdots & \vdots \\ x[N-2] & x[N-3] \end{bmatrix}}_{\mathbf{A} \in \mathbb{R}^{(N-2)\times 2}} \underbrace{\begin{bmatrix} a_1 \\ a_2 \end{bmatrix}}_{\mathbf{a}} = - \underbrace{\begin{bmatrix} x[2] \\ x[3] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\mathbf{b} \in \mathbb{R}^{(N-2)}}.$$

In the noiseless case, two linearly independent equations suffice to solve for (a_1, a_2), so

$$\boxed{N \geq 4}$$

is the smallest N that yields at least two equations ($N - 2 \geq 2$). With noise, take $N > 4$ and solve in least squares (or via SVD).

(c) Show how to recover Ω_0 from the estimated AR coefficients, and then estimate c_0 and ϕ_0 by a linear fit. **[6 Marks]**

Frequency from AR(2). From part (a), $a_1 = -2\cos\Omega_0$ and $a_2 = 1$. Hence

$$\boxed{\widehat{\Omega}_0 \;=\; \arccos\!\big(-\tfrac{a_1}{2}\big), \quad \widehat{a}_2 \approx 1}.$$

(With noise, $\widehat{a}_2$ will deviate from 1; one may optionally project the roots to unit magnitude or average Ω_0 from the angle of the roots.)

Amplitude and phase via linear least squares. Use the identity

$$x[n] \;=\; c_0\cos(\Omega_0 n + \phi_0) \;=\; \alpha\cos(\Omega_0 n) \;-\; \beta\sin(\Omega_0 n),$$

with $\alpha = c_0\cos\phi_0$ and $\beta = c_0\sin\phi_0$. With $\widehat{\Omega}_0$ known, build the regression

$$\underbrace{\begin{bmatrix} \cos(\widehat{\Omega}_0\cdot 0) & -\sin(\widehat{\Omega}_0\cdot 0) \\ \cos(\widehat{\Omega}_0\cdot 1) & -\sin(\widehat{\Omega}_0\cdot 1) \\ \vdots & \vdots \\ \cos(\widehat{\Omega}_0\cdot (N-1)) & -\sin(\widehat{\Omega}_0\cdot (N-1)) \end{bmatrix}}_{\mathbf{G}\in\mathbb{R}^{N\times 2}} \begin{bmatrix}\alpha\\ \beta\end{bmatrix} \;\approx\; \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}.$$

Solve $(\widehat{\alpha}, \widehat{\beta})$ in least squares:

$$[\widehat{\alpha}, \widehat{\beta}]^\top = \arg\min_{\alpha,\beta} \; \|\mathbf{G}[\alpha,\beta]^\top - \mathbf{x}\|_2^2.$$

Finally,

$$\boxed{\widehat{c}_0 \;=\; \sqrt{\widehat{\alpha}^{\,2} + \widehat{\beta}^{\,2}}, \qquad \widehat{\phi}_0 \;=\; \operatorname{atan2}(\widehat{\beta},\, \widehat{\alpha})}.$$

Remarks. (i) This two-step method (AR for frequency, linear LS for amplitude/phase) is robust and fast. (ii) Equivalent complex-exponential LS: fit $x[n] \approx d\, e^{j\widehat{\Omega}_0 n} + \overline{d}\, e^{-j\widehat{\Omega}_0 n}$ to get $d = (c_0/2)e^{j\phi_0}$.

(d) Briefly explain how this extends to K distinct cosines. **[4 Marks]**

Generalization. A sum of K real cosines corresponds to K complex-conjugate pole pairs on the unit circle, i.e., $2K$ complex exponentials. The sequence satisfies an AR($2K$) recursion whose monic polynomial has roots at $e^{\pm j\Omega_k}$:

$$A(z) = \prod_{k=0}^{K-1} \big(1 - e^{j\Omega_k} z^{-1}\big)\big(1 - e^{-j\Omega_k} z^{-1}\big).$$

Estimate AR coefficients from a linear prediction system (with $N \geq 2\cdot(2K)$ recommended in noise), find the roots $\{e^{\pm j\widehat{\Omega}_k}\}$ for frequencies, then solve a linear LS for the amplitudes/phases of each cosine (e.g., using a cosine–sinusoid design matrix per frequency).

Q5 **Sparse Signals and Smooth approximations to ℓ_1 for Denoising** **[20 Marks]**

Context and goal. We observe a noisy vector $\mathbf{g} \in \mathbb{R}^N$ (e.g., wavelet coefficients of a noisy image or audio). We wish to estimate a denoised vector $\mathbf{f}$ whose entries are mostly small or zero (sparse), while staying close to the data. The classical approach uses the ℓ_1 penalty and yields the soft-thresholding rule. Here we study a smooth approximation of the ℓ_1 norm that is easier to optimize with gradient-based methods and to differentiate through (e.g., for learning pipelines), and we examine its shrinkage behavior and its relationship to soft-thresholding.

We consider the separable cost

$$\mathcal{C}_\lambda(\mathbf{f}) = \|\mathbf{g} - \mathbf{f}\|_2^2 \; + \; \lambda \sum_{n=1}^{N} \sqrt{f_n^2 + \varepsilon}, \qquad \lambda > 0, \quad \varepsilon > 0.$$

This replaces the non-differentiable ℓ_1 norm by the smooth surrogate $\sqrt{f^2 + \varepsilon}$. Throughout, notice the problem decouples across coordinates; thus we can analyze the one-dimensional objective

$$\phi(f) \; = \; (g - f)^2 + \lambda \sqrt{f^2 + \varepsilon}$$

and apply the result component-wise.

(a) Explain why the surrogate penalty $\sqrt{f^2 + \varepsilon}$ is a reasonable smooth approximation to $|f|$, and why replacing $|f|$ by it can still promote sparse (or near-sparse) recovery. **[4 Marks]**

Hint. Sketch the graphs of $|f|$ and $\sqrt{f^2 + \varepsilon}$ for a small ε and compare them near the origin.

The function $\sqrt{f^2 + \varepsilon}$ is smooth for all f when $\varepsilon > 0$, and pointwise

$$\sqrt{f^2 + \varepsilon} \; \to \; |f| \quad \text{as } \varepsilon \to 0^+.$$

Hence it closely approximates the ℓ_1 norm while avoiding its non-differentiable kink at $f = 0$. This is useful when one prefers smooth objectives (e.g., for gradient-based solvers or when embedding the estimator as a differentiable layer). The derivative is

$$\frac{d}{df}\sqrt{f^2 + \varepsilon} \; = \; \frac{f}{\sqrt{f^2 + \varepsilon}},$$

which approaches $\operatorname{sgn}(f)$ as $\varepsilon \to 0^+$ for $f \neq 0$. Therefore the penalty retains the same qualitative shrinkage behavior as $|f|$ while remaining smooth.

(b) Differentiate $\phi(f)$ with respect to f and derive the first-order stationarity condition for a minimizer. **[4 Marks]**

Differentiating and equating to zero gives

$$\phi'(f) \; = \; 2(f - g) + \lambda \frac{f}{\sqrt{f^2 + \varepsilon}} \; = \; 0.$$

Equivalently,

$$g \; = \; f\left(1 + \frac{\lambda}{2\sqrt{f^2 + \varepsilon}}\right).$$

(c) Use the rearranged stationarity condition

$$g \; = \; f\left(1 + \frac{\lambda}{2\sqrt{f^2 + \varepsilon}}\right)$$

to show that the minimizer $f^\star$ has the same sign as g, and that its magnitude is no larger than that of g. Show also that $|f^\star| < |g|$ whenever $g \neq 0$. **[4 Marks]**

Since

$$1 + \frac{\lambda}{2\sqrt{f^2 + \varepsilon}} > 1,$$

the multiplicative factor is strictly positive. Hence $f^\star$ and g must have the same sign (or both be zero). Taking absolute values,

$$|g| = |f^\star| \left(1 + \frac{\lambda}{2\sqrt{(f^\star)^2 + \varepsilon}}\right),$$

so

$$|f^\star| = \frac{|g|}{1 + \frac{\lambda}{2\sqrt{(f^\star)^2 + \varepsilon}}} \leq |g|.$$

If $g \neq 0$, then the denominator is strictly greater than 1, so in fact

$$|f^\star| < |g|.$$

Thus the estimator preserves sign and shrinks the magnitude toward zero.

(d) Show that for any finite $\varepsilon > 0$, the minimizer of $\phi(f)$ cannot be $f = 0$ whenever $g \neq 0$. **[4 Marks]**

Hint: Evaluate $\phi'(0)$ and recall that ϕ is convex.

Briefly explain what this implies about the ability of the smooth surrogate to produce exact sparsity compared with the ℓ_1 penalty.

At $f = 0$,

$$\phi'(0) = 2(0 - g) + \lambda \cdot \frac{0}{\sqrt{\varepsilon}} = -2g.$$

If $g \neq 0$, then $\phi'(0) \neq 0$, so $f = 0$ cannot be a stationary point. Also,

$$\phi''(f) = 2 + \lambda \frac{\varepsilon}{(f^2 + \varepsilon)^{3/2}} > 2,$$

so ϕ is strictly convex and has a unique minimizer. Therefore, for any finite $\varepsilon > 0$ and $g \neq 0$, the solution $f^\star$ is nonzero. This means that the smooth surrogate produces shrinkage but not exact sparsity (except when $g = 0$), unlike the ℓ_1 penalty, which can set coefficients exactly to zero.

(e) How is this related to soft-thresholding?

Consider the limit $\varepsilon \to 0^+$ in the first-order condition and recall the soft-threshold operator.

The *soft-threshold* operator with threshold $\tau > 0$ is defined componentwise as

$$\text{soft}_\tau(g) = \begin{cases} \text{sgn}(g)\,(|g| - \tau), & |g| > \tau, \\ 0, & |g| \leq \tau. \end{cases}$$

Show that, in the limit $\varepsilon \to 0^+$, the minimizer approaches a soft-thresholding rule, and determine the relation between τ and λ for the present scaling. **[4 Marks]**

As $\varepsilon \to 0^+$,

$$\frac{f}{\sqrt{f^2 + \varepsilon}} \to \text{sgn}(f) \quad (f \neq 0),$$

and the stationarity condition becomes

$$2(f - g) + \lambda\, \text{sgn}(f) = 0.$$

Solving this gives

$$f^\star = \operatorname{sgn}(g)\ \max\left(|g| - \frac{\lambda}{2},\, 0\right),$$

which is exactly the soft-threshold rule. Therefore, for the present scaling,

$$\tau = \frac{\lambda}{2}.$$

Hence the smooth surrogate can be viewed as a differentiable approximation whose solutions approach soft-thresholding as $\varepsilon \to 0^+$.